CISS245: Advanced Programming
Quiz q3002

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Q1. Write a struct for `Human` so that a `Human` variable has

- `id`: an integer (to identify the human)
- `father`: a pointer to `Human` (NULL if not known)
- `mother`: a pointer to `Human` (NULL if not known)
- `children`: an array to 10 Human pointers
- `children_size`: number of children represented in `children`. If `children_size` is 2, then only `children[0]` and `children[1]` contains meaningful data.

Then create the following;

- 0 is a human. Father and mother unknown. Children are 2 and 3.
- 1 is a human. Father and mother unknown. Children are 2 and 3.
- 2 is a human. Father is 0 and mother is 1. No children.
- 3 is a human. Father is 0 and mother is 1. Child is 4.
- 4 is a human. Father is 3 and mother is unknown. No children.

Answer:

```
// create human struct here
#include <iostream>

struct Human
{
  int id
  Human * father;
  Human * mother;
  Human * children[10];
  int children_size;
}

// create humans 0,1,2,3,4 here

int main()
{
  Human h0, h1, h2, h3, h4;

  h0.id = 0;
  h0.father = NULL;
  h0.mother = NULL;
```

```
  h0.children[0] = &h2;
  h0.children[1] = &h3;
  h0.children_size = 2;

  h1.id = 1;
  h1.father = NULL;
  h1.mother = NULL;
  h1.children[0] = &h2;
  h1.children[1] = &h3;
  h1.children_size = 2;

  h2.id = 2;
  h2.father = &h0;
  h2.mother = &h1;
  h2.children_size = 0;

  h3.id = 3;
  h3.father = &h0;
  h3.mother = &h1;
  h3.children[0] = &h4;
  h3.children_size = 1;

  h4.id = 4;
  h4.father = &h3;
  h4.mother = NULL;
  h4.children_size = 0;

  return 0;
}
```

Instructions

In the file `thispreamble.tex` look for

```
\renewcommand\AUTHOR{}
```

and enter your email address:

```
\renewcommand\AUTHOR{jdoe5@cougars.ccis.edu}
```

(This is not really necessary since alex will change that for you when you execute `make`.) In your bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

Enter your answers in `main.tex`. In the bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

For each question, you'll see boxes for you to fill. For small boxes, if you see

```
1 + 1 = \answerbox{}.
```

you do this:

```
1 + 1 = \answerbox{2}.
```

`answerbox` will also appear in "true/false" and "multiple-choice" questions.

For longer answers that need typewriter font, if you see

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
\end{answercode}
```

you do this:

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
int x;
\end{answercode}
```

`answercode` will appear in questions asking for code, algorithm, and program output. In this case, indentation and spacing is significant. For program output, I do look at spaces and newlines.

For long answers (not in typewriter font) if you see

```
What is the color of the sky?
\begin{answerlong}
\end{answerlong}
```

you can write

```
What is the color of the sky?
\begin{answerlong}
The color of the sky is blue.
\end{answerlong}
```

A question that begins with "T or F or M" requires you to identify whether it is true or false, or meaningless. "Meaningless" means something's wrong with the question and it is not well-defined. Something like "$1 + 2 = 4$" is either true or false (of course it's false). Something like "$1+_2 = 4$?" does not make sense.

When writing results of computations, make sure it's simplified. For instance write 2 instead of $1 + 1$.

Higher level classes.

For students beyond 245: You can put LaTeX commands in `answerlong`.

More examples of meaningless statements: Questions such as "Is $42 = 1+_2$ true or false?" or "Is $42 = \{2\}^{\{3\}}$ true or false?" does not make sense. "Is $P(42) = \{42\}$ true or false?" is meaningless because $P(X)$ is only defined if X is a set. For "Is $1 + 2 + 3$ true or false?", "$1 + 2 + 3$" is well–defined but as a "numerical expression", not as a "proposition", i.e., it cannot be true or false. Therefore "Is $1 + 2 + 3$ true or false?" is also not a well-defined question.

More examples of simplification: When you write down sets, if the answer is $\{1\}$, do not write $\{1, 1\}$. And when the values can be ordered, write the elements of the set in ascending order. When writing polynomials, begin with the highest degree term.

When writing a counterexample, always write the simplest.